Abstract Linear Algebra

Fisher Yu

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Problem 1: The First Isomorphism Theorem for Vector Spaces

Let V and W be $\mathbb{F}$-vector spaces and let $T : V \to W$ be linear. Define

$$\varphi : V/\ker(T) \to W, \qquad \varphi(v + \ker(T)) = T(v).$$

(a) φ is well-defined

Suppose $v_1 + \ker(T) = v_2 + \ker(T)$. We must show $T(v_1) = T(v_2)$.

Since $v_1 + \ker(T) = v_2 + \ker(T)$ if and only if $v_1 - v_2 \in \ker(T)$. So $T(v_1 - v_2) = 0$. Since T is linear,

$$T(v_1) - T(v_2) = T(v_1 - v_2) = 0,$$

so $T(v_1) = T(v_2)$. Such that is is well-defined. ■

(b) φ is linear

Let $v_1 + \ker(T), v_2 + \ker(T) \in V/\ker(T)$ and $\lambda \in \mathbb{F}$.

Additivity. By definition of addition in the quotient space,

$$\varphi\big((v_1 + \ker(T)) + (v_2 + \ker(T))\big) = \varphi\big((v_1 + v_2) + \ker(T)\big) = T(v_1 + v_2) = T(v_1) + T(v_2),$$

by def this equals $\varphi(v_1 + \ker(T)) + \varphi(v_2 + \ker(T))$.

Homogeneity. Similarly,

$$\varphi\big(\lambda(v_1 + \ker(T))\big) = \varphi(\lambda v_1 + \ker(T)) = T(\lambda v_1) = \lambda T(v_1) = \lambda\, \varphi(v_1 + \ker(T)).$$

Hence φ is linear. ■

(c) φ is injective

It suffices to show $\ker(\varphi) = \{0 + \ker(T)\}$, the zero element of $V/\ker(T)$.

Suppose $\varphi(v+\ker(T)) = 0$, i.e., $T(v) = 0$. Then $v \in \ker(T)$, which means $v+\ker(T) = 0 + \ker(T)$. Thus $\ker(\varphi) = \{0 + \ker(T)\}$, and since φ is linear, this shows φ is injective (a linear map is injective iff its kernel is trivial). ■

(d) $\mathrm{Im}(\varphi) = \mathrm{Im}(T)$

($\subseteq$) If $w \in \mathrm{Im}(\varphi)$, then $w = \varphi(v + \ker(T)) = T(v)$ for some $v \in V$, so $w \in \mathrm{Im}(T)$.

($\supseteq$) If $w \in \mathrm{Im}(T)$, then $w = T(v)$ for some $v \in V$. Then $w = T(v) = \varphi(v + \ker(T))$, so $w \in \mathrm{Im}(\varphi)$.

Hence $\mathrm{Im}(\varphi) = \mathrm{Im}(T)$. ■

(e) Conclusion

By (b), φ is linear. By (c), φ is injective. By (d), φ is surjective onto $\mathrm{Im}(T)$ (viewing φ as a map $V/\ker(T) \to \mathrm{Im}(T)$). Therefore

$$\varphi : V/\ker(T) \to \mathrm{Im}(T)$$

is a linear bijection, i.e., an isomorphism. We conclude

$$V/\ker(T) \;\cong\; \mathrm{Im}(T). \qquad \blacksquare$$

Problem 2

(a)

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be rotation by 90° counterclockwise about the origin. With respect to the standard basis $\{e_1, e_2\}$ of $\mathbb{R}^2$, we have $T(e_1) = e_2$ and $T(e_2) = -e_1$, so the matrix of T is

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The eigenvalues of T are the roots (in $\mathbb{R}$) of the characteristic polynomial

$$\det(A - \lambda I) = \det \begin{pmatrix} -\lambda & -1 \\ 1 & -\lambda \end{pmatrix} = \lambda^2 + 1.$$

We claim $\lambda^2 + 1 = 0$ has no solution in $\mathbb{R}$. Indeed, for any $\lambda \in \mathbb{R}$, $\lambda^2 \geq 0$, so $\lambda^2 + 1 \geq 1 > 0$. Hence there is no real λ with $\lambda^2 + 1 = 0$, so A (and hence T) has no real eigenvalues. ■

(i.e consider eigenvector is a kind of vectors which only change "size" in linear transformation, so that it is easy to find out in the operation of rotation in $\mathbb{R}^2$, all the vectors except zero should be rotated, so it make sense that the matrix have no eigenvalue.)

(b)

Let $T : \mathbb{C} \to \mathbb{C}$ be given by $T(z) = iz$. We claim i is an eigenvalue of T.

Take any nonzero vector, e.g., $z = 1 \in \mathbb{C}$ (viewing $\mathbb{C}$ as a 1-dimensional $\mathbb{C}$-vector space over itself). Then

$$T(1) = i \cdot 1 = i = i \cdot 1,$$

so $T(1) = i \cdot 1$ with $1 \neq 0$. By definition, this means 1 is an eigenvector of T with eigenvalue $i \in \mathbb{C}$. Hence T has an eigenvalue. ■

(i.e In fact, since $\mathbb{C}$ is 1-dimensional over itself, every nonzero $z \in \mathbb{C}$ satisfies $T(z) = iz$, so every nonzero vector is an eigenvector with eigenvalue i.)

(c)

In part (a), $V = \mathbb{R}^2$ is a 2-dimensional vector space over $\mathbb{F} = \mathbb{R}$, and scalars are real numbers. Multiplying a vector in $\mathbb{R}^2$ by a real scalar λ only *scales* the vector (stretches, shrinks, or reverses it) — it never rotates it. Since a 90° rotation moves every nonzero vector off of its own line (span), no nonzero vector can satisfy $T(v) = \lambda v$ for real λ, so there are no real eigenvalues.

In part (b), $V = \mathbb{C}$ is a 1-dimensional vector space over $\mathbb{F} = \mathbb{C}$ itself, and scalars are *complex* numbers. Multiplication by a complex number i is not merely scaling: geometrically, multiplying by $i = e^{i\pi/2}$ rotates the complex plane by 90°. So the "scalar multiplication" $z \mapsto iz$ *is itself* the 90° rotation. Consequently, the rotation map T coincides exactly with scalar multiplication by i, so every nonzero vector is trivially an eigenvector with eigenvalue i.

The deeper point: whether a rotation has eigenvalues depends on whether the field of scalars is rich enough to express that rotation as multiplication by a scalar. Over $\mathbb{R}$, scalars can only stretch/shrink/flip a line, so a rotation (which moves vectors off their

span) can never be scalar multiplication. Over $\mathbb{C}$, scalars can rotate (since complex numbers have both magnitude and argument), so a rotation *can* be realized as multiplication by a single complex scalar. ■

Check-In

A little recall to the Problem1, since vector space also construct a abelian group structure, the first isomorphic theorem is pretty much the same in the group theory. This reminds me of these kind of things: quotient space, coset, ideal, are they essentially the same or the same idea? What is the motivation of this idea? And also I just started learing category theory, how can we view these perspectives from the standpoint of category theory?

For my favourite quote, which inspire me very much was from the game league of legend: A true master is an eternal student–master Yi